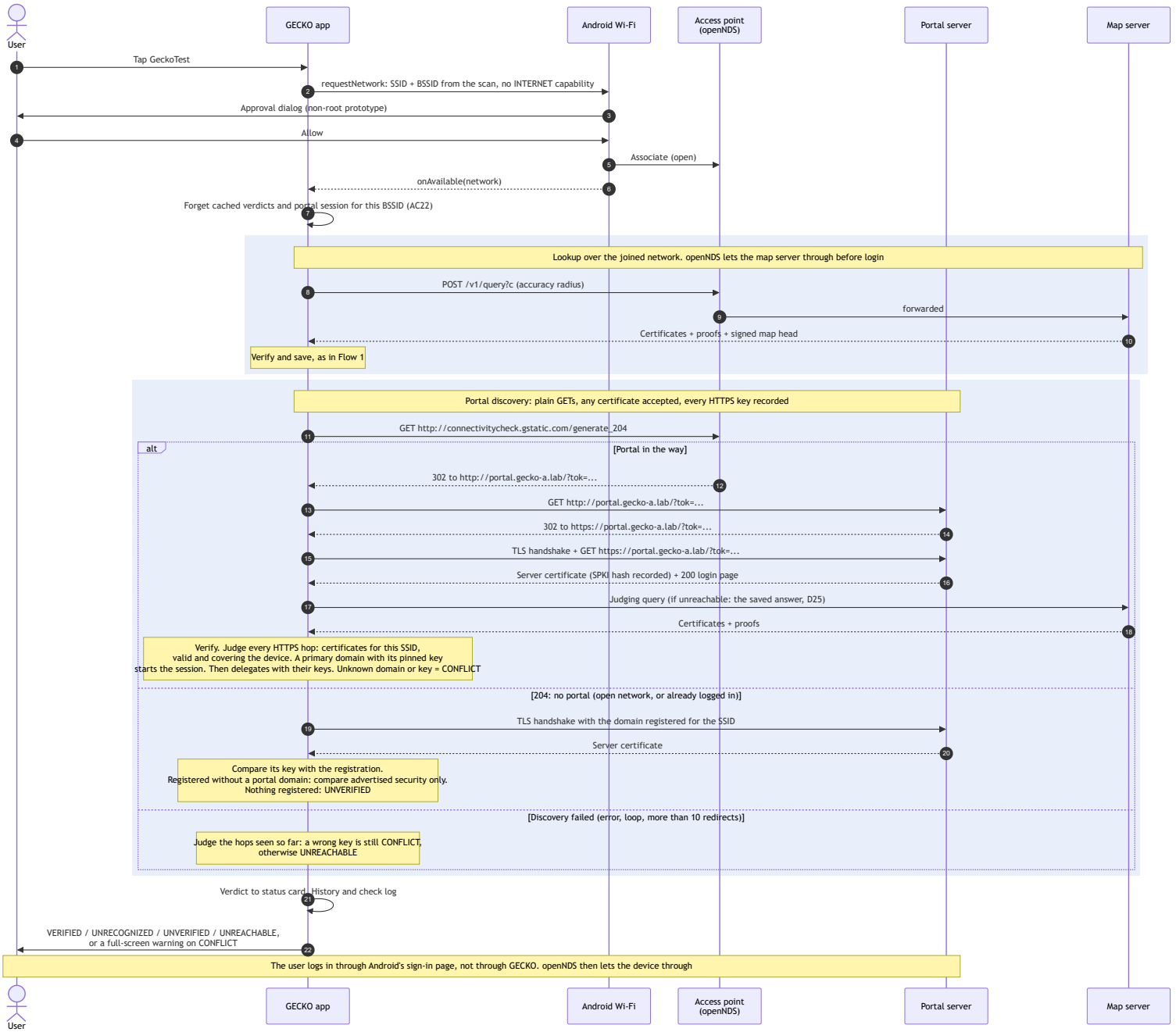

User
GECKO app
Android Wi-Fi
Access point (openNDS)
Portal server
Map server
1
Tap GeckoTest
2
requestNetwork: SSID + BSSID from the scan, no INTERNET capability
3
Approval dialog (non-root prototype)
4
Allow
5
Associate (open)
6
onAvailable(network)
7
Forget cached verdicts and portal session for this BSSID (AC22)
Lookup over the joined network. openNDS lets the map server through before login
8
POST /v1/query?c (accuracy radius)
9
forwarded
10
Certificates + proofs + signed map head
Verify and save, as in Flow 1
Portal discovery: plain GETs, any certificate accepted, every HTTPS key recorded
11
GET http://connectivitycheck.gstatic.com/generate_204
alt
[Portal in the way]
12
302 to http://portal.gecko-a.lab/?tok=...
13
GET http://portal.gecko-a.lab/?tok=...
14
302 to https://portal.gecko-a.lab/?tok=...
15
TLS handshake + GET https://portal.gecko-a.lab/?tok=...
16
Server certificate (SPKI hash recorded) + 200 login page
17
Judging query (if unreachable: the saved answer, D25)
18
Certificates + proofs
Verify. Judge every HTTPS hop: certificates for this SSID,
valid and covering the device. A primary domain with its pinned key
starts the session. Then delegates with their keys. Unknown domain or key = CONFLICT
[204: no portal (open network, or already logged in)]
19
TLS handshake with the domain registered for the SSID
20
Server certificate
Compare its key with the registration.
Registered without a portal domain: compare advertised security only.
Nothing registered: UNVERIFIED
[Discovery failed (error, loop, more than 10 redirects)]
Judge the hops seen so far: a wrong key is still CONFLICT,
otherwise UNREACHABLE
21
Verdict to status card, History and check log
22
VERIFIED / UNRECOGNIZED / UNVERIFIED / UNREACHABLE,
or a full-screen warning on CONFLICT
The user logs in through Android's sign-in page, not through GECKO. openNDS then lets the device through
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